

Account Opening Concierge Agent

An advisor-facing console. The advisor talks or pastes notes; the account opening form fills itself. Every value the agent produces is shown to the advisor for confirmation before it is saved.

Note: Items highlighted in Green are in scope for the initial prototype.

Who uses it	A financial advisor, or their client-service associate
Shape	Form on the left, chat panel on the right, one screen
Feature areas	Conversational Intake · Adapt and Resume · Chase and Follow-ups · Document Uploads
Data	32 attributes · 12 groups · 5 account registrations [Prototype 10-12 Attributes, 3 Groups, 3 From Types]
Build window	2 weeks, Backend First
Grounding	Citation-based responses to source data where applicable

Conventions

How to read this document

Every feature section is laid out the same way, so you only learn the shape once:

- What it does — one short paragraph.
- Sub-features — each with a sample conversation and a mockup.
- Build notes — complexity, what is in the minimum build, what you need before starting.

The agent's reply format

Every agent reply has up to three parts. Parts 2 and 3 appear only when relevant.

Part	What it is	Always?
Reply	Plain language with Citations for source data; Human Approvals	Yes
Citation line	Where the answer came from: a CRM note, an uploaded document, a rule, or the advisor's own words. Rendered under the reply in smaller grey text.	When grounded
Proposal Approval	The exact fields the agent wants to write, with Confirm / Edit / Reject. Nothing is saved until the advisor acts on it.	When writing fields

Speaker labels used throughout

ADVISOR

What the advisor types or says.

CONCIERGE

What the agent replies.

↳ Where this came from

THE RULE EVERYTHING ELSE FOLLOWS FROM

The agent may reason freely to decide what to ask next. It may never write a value to the record without the advisor confirming it. A wrong guess about the next question costs a second; a wrong value on an account form costs a rejected packet.

Feature index

§	Feature area	Sub-features	Build order
1	Conversational Intake	CRM-assisted start · notes extraction · clarifying questions · correction and history	1
2	Adapt and Resume	Conditional sections · in-conversation validation · session resume	2
3	Chase and Follow-ups	Pre-drafted messages; follow-ups from existing inflight client info; follow-ups from previous interactions; Goal – Keep engaged and momentum going [Prototype Total 3-5 message pools]	1
4	Document Uploads	Documents uploaded to the agent get tagged accordingly to the appropriate placeholders. Good to have: Document Intelligence; extraction	3

1 • Conversational Intake

The advisor types or pastes what they know. The agent works out which form fields that maps to, and shows them for confirmation before anything is saved.

1.1 • Starting from the CRM

The agent can begin on its own by reading the linked contact record and the most recent meeting note, so the advisor starts from a part-filled form rather than a blank one.

CONCIERGE

I pulled the contact record and the meeting note from 12 September. Three things look usable. Want me to carry them over?

↳ Sources · CRM contact (updated 3 days ago) · Meeting note 2026-09-12

ADVISOR

Yes, go ahead.

CONCIERGE

Done — name, email and mobile are in. They are marked as coming from the CRM so you can see where they came from. What kind of account are we opening?

↳ 3 fields filled · none material, so no confirmation needed

New Account · Concierge

Case CASE-0001

Starting from CRM. The agent read the linked contact record and the latest meeting note. Nothing has been saved yet.

ACCOUNT SETUP

Account type

Not set yet

Advisory program

Not set yet

APPLICANT

Legal name

OWNER_1

from CRM

Email

o***1@example.com

from CRM

Mobile

(...) ---4417

from CRM

CONCIERGE

I pulled the contact record and the meeting note from 12 September. Three things look usable. Want me to carry them over?

Sources · CRM contact (updated 3 days ago) · Meeting note 2026-09-12

ADVISOR

Yes, go ahead.

CONCIERGE

Done — name, email and mobile are in. They are marked as coming from the CRM so you can see where they came from. What kind of account are we opening?

Type a reply, or paste your meeting notes...

Submit blocked · 8 fields still empty

Mockup 1.1 — the agent opens the case from CRM data. Purple markers show values that came from the CRM rather than being typed.

Which CRM values need confirming

Value type	Rule	Reason
Material — suitability, titling, beneficiaries, legal address	Always confirm, however fresh	These drive a regulatory determination.
Non-material, updated within 90 days	Accept automatically, show the source	Low consequence if wrong, easily corrected.
Non-material, older than 90 days	Show it, flag the age, one-click accept	Cheap safeguard against carrying forward an old address.

Two separate states in the data so the difference is explicit rather than implied:

```
prefilled_accepted    // carried over automatically, source shown, no click needed
prefilled_confirmed  // advisor explicitly confirmed it – material values only
```

1.2 · Pasting meeting notes

The advisor can paste a paragraph and let the agent take it apart. Several fields can come from one sentence. Everything found is listed with the words it came from, and nothing is written until the advisor confirms.

ADVISOR

They want to open it jointly with my wife. I have 3 kids, one under 18. About 400k, moving it over from somewhere else.

CONCIERGE

Here is what I took from that. Nothing is saved until you confirm.

↳ 5 values proposed • each shows the words it came from

AGENT PROPOSES • NOTHING SAVED UNTIL THE ADVISOR CONFIRMS

account.type **Joint with right of survivorship**
household.dependants **3**
household.minor_dependants **1**
funding.method **Transfer from another firm**
funding.initial_amount **\$400,000**
[Confirm all] [Edit] [Reject]

Note what the agent did not do: it heard nothing about ages or dates of birth and wrote nothing about them.

New Account • Concierge

Case CASE-0001

ACCOUNT SETUP

Account type

Joint with right of survivorship
from: "open it jointly with my wife" confirm

Advisory program

Advisor-Driven typed

HOUSEHOLD

Dependants

3
from: "I have 3 kids" confirm

Minor dependants

1
from: "one under 18" confirm

FUNDING

Funding method

Transfer from another firm
from: "moving it over from somewhere else" confirm

Initial amount

\$400,000
from: "about 400k" confirm

ADVISOR

They want to open it jointly with my wife. I have 3 kids, one under 18. About 400k, moving it over from somewhere else.

CONCIERGE

Here is what I took from that. Nothing is saved until you confirm.

PROPOSED • CONFIRM BEFORE SAVING

account.type **Joint with right of survivorship**
household.dependants **3**
household.minor_dependants **1**
funding.method **Transfer from another firm**
funding.initial_amount **\$400,000**
Confirm all Edit Reject

Type a reply...

Mockup 1.2 — the confirmation ledger. Amber fields are proposed, not saved; each shows the advisor's own words underneath. Submit stays blocked.

1.3 • Asking when a remark is suggestive but not conclusive

Some statements hint at an answer without settling it. The agent holds the hint as context, writes nothing, and asks a one-line question. Filing taxes jointly and holding an account jointly are different things, and account titling is expensive to unwind after funding.

ADVISOR

My wife handles the taxes.

CONCIERGE

Noted. Quick one so I set this up right — is this account just for you, or joint with your spouse? The two are paperworked differently.

↳ Heard: married. Not enough on its own to set the registration.

ADVISOR

Joint.

CONCIERGE

Thanks — that one I will set once you confirm below.

↳ Now explicit · 1 value proposed

The same pattern applies wherever confidence falls below the floor: ask rather than propose.

New Account · Concierge

Case CASE-0001

ACCOUNT SETUP

Account type

Not set yet

The agent heard something suggestive but not conclusive. It will not set this field on an indirect remark.

HELD FOR CONTEXT

Marital status (context only)

Likely married

heard, not written to any form field

context

ADVISOR

My wife handles the taxes.

CONCIERGE

Noted. Quick one so I set this up right — is this account just for you, or joint with your spouse? The two are paperworked differently.

Heard: married. Not enough on its own to set the registration.

ADVISOR

Joint.

CONCIERGE

Thanks — that one I will set once you confirm below.

PROPOSED · CONFIRM BEFORE SAVING

account.type Joint with right of survivorship

Confirm all Edit Reject

Type a reply...

Mockup 1.3 — a suggestive remark held as context only, with the account type left unset until the advisor answers directly.

1.4 · Correcting a proposed value

The advisor can edit any proposed value before confirming it. Both the heard value and the correction are retained. Where a correction creates a conflict elsewhere, the agent says so.

CONCIERGE

I had \$400,000 from the notes. You changed it to \$450,000 — I have kept both, so the record shows what I heard and what you corrected.

↳ Edited before saving · original retained

ADVISOR

Right, they added to it after we spoke.

CONCIERGE

Makes sense. One flag: \$450,000 sits above the band recorded for liquid net worth. Worth revisiting that answer before submission.

↳ Cross-check · funding amount vs declared net worth band

New Account · Concierge

Case CASE-0001

FUNDING

Initial amount

\$450,000

confirmed

Edited by the advisor before confirming. The original heard value and the correction are both kept on the record.

CHANGE HISTORY

PROPOSED

\$400,000 — from “about 400k”

CORRECTED

\$450,000 — advisor edit, 10:48

CONCIERGE

I had \$400,000 from the notes. You changed it to \$450,000 — I have kept both, so the record shows what I heard and what you corrected.

Edited before saving · original retained

ADVISOR

Right, they added to it after we spoke.

CONCIERGE

Makes sense. One flag: \$450,000 sits above the band recorded for liquid net worth. Worth revisiting that answer before submission.

Cross-check · funding amount vs declared net worth band

Type a reply...

Submit blocked · 6 required fields empty

Mockup 1.4 — an advisor correction, with both values kept and the downstream conflict surfaced immediately.

Build notes

Complexity	Medium. The screen is simple; extraction accuracy and the confirmation loop are the real work.
Build order	1 of 4 — everything else assumes this exists.
Minimum build	Free text and paste input · extraction limited to the 14 attributes marked inferable · every proposed value carries the words it came from · Confirm, Edit and Reject per value plus Confirm all · below the confidence floor the agent asks instead of proposing · edits retain the original.
Before you start	The attribute catalog (section 5) · a confidence floor, start at 0.70 · roughly 20 real advisor notes to test extraction against · read access to CRM contact and meeting-note records.

2 · Adapt and Resume

The form changes shape as the conversation goes on, validates as it goes, and picks up exactly where it was left.

2.1 · Opening and closing sections

Some answers pull in a whole extra block of questions — a delivering institution, a joint owner, a trustee. The agent opens those, closes the ones that no longer apply, and says why. Where an answer has a consequence the advisor should know about now, it says so immediately.

ADVISOR

There is an institution holding the funds right now.

CONCIERGE

Understood — that makes this an incoming transfer, so I have opened a short section for the delivering firm.
 Worth knowing now: the registration on their side has to match the registration here. If it does not, the transfer bounces and you lose about a week.

↳ Basis • transfer registration-match rule • 4 fields added, 1 section closed

New Account • Concierge

Case CASE-0001

FUNDING

Funding method

Transfer from another firm

confirmed

DELIVERING INSTITUTION

OPENED BY YOUR ANSWER

Institution name

Not set yet

Their account number

Not set yet

Registration on that account

Not set yet

Registration ID

Not set yet

Employment section was closed – it does not apply to this registration.

ADVISOR

There is an institution holding the funds right now.

CONCIERGE

Understood — that makes this an incoming transfer, so I have opened a short section for the delivering firm.

Worth knowing now:

the registration on their side has to match the registration here. If it does not, the transfer bounces and you lose about a week.

Basis • transfer registration-match rule • 4 fields added, 1 section closed

ADVISOR

Good to know. I will get the details.

Type a reply...

Submit blocked • 4 required fields empty

Mockup 2.1 — a new section appears, an irrelevant one closes, and the consequence is explained in the same breath.

Section triggers

When this is answered	This opens	This closes
Account type = Joint	Joint owner block (1–3 owners)	—
Account type = Trust or UTMA	Entity block: title, responsible parties, certification	Joint owner block
Account type = IRA	Beneficiaries, with percentage validation	Joint owner block
Account type = Trust or UTMA	—	Employment (does not apply to an entity owner)
Funding = transfer or rollover	Delivering institution: name, account, registration, ID	—
Employment = employed or self-employed	Employer name	—
FINRA affiliation = yes	Approval letter upload — blocks submission until attached	—

2.2 · Validation inside the conversation

Checks run as values arrive rather than at the end, so the advisor hears about a problem while the client is still on the call.

ADVISOR

Two beneficiaries, 60 and 30.

CONCIERGE

That comes to 90%. Beneficiary shares have to total exactly 100, so there are 10 points unassigned. Add a third person, or adjust the two you have?

↳ Checked as entered · 90 of 100 allocated

ADVISOR

Make the second one 40.

CONCIERGE

That balances. Confirm the change and I will save it.

↳ Recheck · 100 of 100 allocated

New Account · Concierge

Case CASE-0001

BENEFICIARIES

Beneficiary 1 · share

60%

confirmed

Beneficiary 2 · share

30%

confirmed

Total allocated

90% — must be exactly 100%

Submission stays blocked while any percentage set is short or over.

ADVISOR

Two beneficiaries, 60 and 30.

CONCIERGE

That comes to 90%. Beneficiary shares have to total exactly 100, so there are 10 points unassigned. Add a third person, or adjust the two you have?

Checked as entered · 90 of 100 allocated

ADVISOR

Make the second one 40.

CONCIERGE

That balances. Confirm the change and I will save it.

Recheck · 100 of 100 allocated

Type a reply...

Submit blocked · allocation incomplete

Mockup 2.2 — a percentage set that does not balance, caught as it is entered rather than at submission.

The three kinds of check

Kind	Example	When it runs
Is the value sensible on its own	A date that exists · a percentage between 0 and 100 · a nine-digit tax identifier	As the value is entered
Does it agree with what is already there	Beneficiary shares totalling 100 · funding amount against declared net worth · a name matching across two documents	On every change to either side

Does it trigger a rule worth mentioning now	Transfer registration match · every acting trustee must sign · approval letter required	When the triggering value is confirmed
---	---	--

2.3 · Coming back to it

Onboarding is almost never finished in one sitting. When the advisor returns, the agent says where things stopped and what is left, rather than reopening a form at the top.

CONCIERGE

Welcome back. You stopped on Tuesday partway through the delivering institution. Two things left there: their account number and the registration ID. Want to carry on from there, or jump somewhere else?

↳ Session restored · 12 fields held · nothing expired

New Account · Concierge

Resumed · last active 2 days ago

Picking up where you stopped. You were on the delivering institution section on Tuesday. Nothing was lost.

WHERE

THINGS

STAND

DONE

Account setup · 3 of 3

DONE

Applicant details · 7 of 7

PARTIAL

Delivering institution · 2 of 4

NOT STARTED

Trusted contact · 0 of 4

NOT STARTED

Disclosures · 0 of 2

CONCIERGE

Welcome back. You stopped on Tuesday partway through the delivering institution. Two things left there: their account number and the registration ID.

Want to carry on from there, or jump somewhere else?

Session restored · 12 fields held · nothing expired

Type a reply...

Submit blocked · 10 required fields empty

Mockup 2.3 — resuming after two days, with a plain-language summary of what is done and what is not.

Build notes

Complexity	Low, deliberately. This is rules, not reasoning, and can be tested without a model in the loop.
Build order	2 of 4.
Minimum build	Five registrations · the seven section triggers above · live count of what is left · all three kinds of check running in the conversation · full session resume with nothing expiring.
Before you start	The question list with its dependencies · the rules that open and close each section · a test that proves every required field is reachable on every path.

3 · Chase and Follow-ups

Most of the elapsed time in opening an account is waiting. This section makes the waiting shorter, the asking smarter, and the whole book visible at once.

3.1 · Ordered by what is actually blocking

Outstanding items are ranked by whether they stop submission, not by where they sit on the form. Three missing fields where one blocks and two do not is a very different situation from three that all block.

CONCIERGE

Three things are still open on this one, and one of them is blocking submission on its own. Before you chase it — you will need the registration ID for the third-party institution. Easier to collect that in the same call than to come back for it.

↳ Blocking · 1 of 3 · oldest item 5 days

ADVISOR

Send them a note listing everything.

CONCIERGE

Drafted, in your voice. Have a read before it goes.

↳ Draft only · nothing sends without your approval

New Account · Concierge

Follow-up queue

WHAT IS HOLDING THIS UP

Registration ID · delivering institution
Needed before the transfer can be requested. The advisor has been asked twice; last reminder went out yesterday morning.

Trustee certification page
Not yet uploaded. This blocks submission on its own.

Trusted contact
Can be declined, but the decline has to be recorded. Not answered yet.

REMINDER SCHEDULE

SENT Day 2 · gentle nudge

SENT Day 4 · listed what is missing

QUEUED Day 7 · flag to the advisor instead

CONCIERGE

Three things are still open on this one, and one of them is blocking submission on its own.

Before you chase it — you will need the registration ID for the third-party institution. Easier to collect that in the same call than to come back for it.

Blocking · 1 of 3 · oldest item 5 days

ADVISOR

Send them a note listing everything.

CONCIERGE

Drafted, in your voice. Have a read before it goes.

Type a reply...

Submit blocked · 1 blocking item

Mockup 3.1 — outstanding items ordered by blocking impact, with the reminder schedule visible and the next nudge queued.

3.6 · Dormancy and revival

Cases go quiet. Rather than chasing indefinitely or letting them rot, the agent marks a case dormant after fourteen days without movement and proposes one of three actions.

Action	What happens	When it fits
Restart	A fresh link is issued; everything already confirmed is retained	The client is still interested but lost the thread
Close out	The case is archived with its reason; the record is retained	The client has gone elsewhere or changed their mind
Hand to the advisor	Chasing stops entirely; the advisor takes it offline	The relationship needs a phone call, not another message

4 • Document Uploads

Documents attach inside the conversation, and the agent reads each one as it arrives.

4.1 • Checking a document the moment it lands

The point of reading on arrival is timing. The same problem found three days later by somebody else costs a phone call and a week; found now, it costs one sentence.

ADVISOR

[attached] licence-photo.jpg

CONCIERGE

Got it, but that is the front only — I need the back as well. A phone photo is fine, same as the first one.

↳ Checked • readable • not expired • back side missing

ADVISOR

[attached] statement-jan.pdf

CONCIERGE

This one is from January, and proof of address has to be within 90 days. Anything more recent will do.

↳ Checked • document is 8 months old • limit is 90 days

New Account • Concierge

Case CASE-0001

DOCUMENTS

Photo ID • front

Accepted • read on arrival

Photo ID • back

Still needed

Proof of address

Accepted • dated within 90 days

Trustee certification

Still needed • blocks submission

Drag a file here, or attach it in the chat

ADVISOR

[attached] licence-photo.jpg

CONCIERGE

Got it, but that is the front only — I need the back as well. A phone photo is fine, same as the first one.

Checked • readable • not expired • back side missing

ADVISOR

[attached] statement-jan.pdf

CONCIERGE

This one is from January, and proof of address has to be within 90 days. Anything more recent will do.

Checked • document is 8 months old • limit is 90 days

Type a reply, or attach a file...

Submit blocked • 2 documents outstanding

Mockup 4.1 — documents attached in the chat, each checked on arrival with the specific problem named rather than a generic rejection.

What is checked on every upload

Check	What it catches
-------	-----------------

Completeness	Front-only identity documents · missing pages · a signature block left blank
Readability	Photographs too dark, blurred or cropped to be read
Age	Proof of address older than 90 days · an expired identity document
Right document	A tax form submitted where a different one was asked for
Name match	The name on the document differs from the name on the application
Title match	A trust certification whose registered title differs from the one entered

4.2 · Reading values out of a document

Where a document contains values the form needs, the agent reads them and proposes them through the same confirmation ledger used for conversation.

ADVISOR

[attached] transfer-statement.pdf

CONCIERGE

I read four things off page 1. Check them against the document before you confirm — I can misread a scan. One thing lines up well: the registration on their side matches the one here, so the transfer should not bounce on that.

↳ Source · uploaded statement, page 1 · registration-match check passed

AGENT PROPOSES · NOTHING SAVED UNTIL THE ADVISOR CONFIRMS

funding.delivering_institution.name **THIRD_PARTY_BROKER**
funding.delivering_institution.account_no **●●●●-●●8832**
funding.delivering_institution.registration **Joint WROS**
funding.delivering_institution.value **\$402,118.44**
[Confirm all] [Edit] [Reject]

New Account · Concierge

Case CASE-0001

DELIVERING INSTITUTION

Institution name

THIRD_PARTY_BROKER

confirm

read from the uploaded statement, page 1

Their account number

••••••8832

confirm

read from the uploaded statement, page 1

Registration on that account

Joint with right of survivorship

confirm

read from the uploaded statement, page 1

Account value shown

\$402,118.44

confirm

read from the uploaded statement, page 1

ADVISOR

[attached] transfer-statement.pdf

CONCIERGE

I read four things off page 1. Check them against the document before you confirm — I can misread a scan.

One thing lines up well:

the registration on their side matches the one here, so the transfer should not bounce on that.

Source · uploaded statement, page 1 · registration-match check passed

PROPOSED · CONFIRM BEFORE SAVING

funding.delivering_institution.name

THIRD_PARTY_BROKER

funding.delivering_institution.account_no

••••••8832

funding.delivering_institution.registration

Joint WROS

funding.delivering_institution.value

\$402,118.44

Confirm all

Edit

Reject

Type a reply...

Mockup 4.2 — values read from an uploaded statement, each citing the page it came from, all awaiting confirmation.

4.3 · Where the agent stops

Two behaviours, both worth demonstrating: handing an advice question back, and refusing to invent a value even when instructed to.

ADVISOR

Should they go with the growth model or the income one?

CONCIERGE

That is an advice question, so I am handing it to you. I have flagged it on the case so it is on the record.

↳ Escalated · advice · logged 10:42

ADVISOR

Fine. Just put whatever you think for the income band.

CONCIERGE

I cannot guess that one — it feeds the suitability review. Give me a band and I will set it.

↳ Refused · value would reach the packet unverified

New Account · Concierge

Case CASE-0001

INVESTMENT PROFILE

Risk tolerance

Moderate

confirmed

Objective

Growth

confirmed

Annual income

Not set yet

IDENTITY

SSN / TIN

.....-1234

held securely

This value was typed straight into a secure field. The agent never receives it.

ADVISOR

Should they go with the growth model or the income one?

CONCIERGE

That is an advice question, so I am handing it to you. I have flagged it on the case so it is on the record.

Escalated · advice · logged 10:42

ADVISOR

Fine. Just put whatever you think for the income band.

CONCIERGE

I cannot guess that one — it feeds the suitability review. Give me a band and I will set it.

Refused · value would reach the packet unverified

Type a reply...

Submit blocked · 1 required field empty

Mockup 4.3 — the agent declining to advise and declining to guess. The identity field on the left was typed into a secure field the agent never sees.

Build notes

Complexity	Medium to high. The checks matrix is more work than the reading itself.
Build order	3 of 4.
Minimum build	Four document types · all six checks above · phone photographs accepted · value extraction from the transfer statement · rejected uploads kept with their reason · advice escalation and refusal-to-guess.
Before you start	Which documents are required for each registration · how recent each must be · roughly 40 sample documents including deliberately defective ones.

5 · Citation format and retrieval approach

The agent needs session state to reason. It must know that annual income is filled and confirmed to decide what to ask next. It does not need the advisor's phone number to do that. A deterministic redaction layer strips PII before the session state enters the prompt. It is not model-powered. It reads the PII column from the attribute schema and applies the rule mechanically. No PII value ever reaches the LLM endpoint.

PII class	Fields (examples)	What the LLM receives	Example in prompt
none	account.type, risk_tolerance, disclosures, funding.method	Full value	"risk_tolerance": "MODERATE"
indirect	citizenship_status, employment.status, income band, net worth band	Full value (categorical, not identifying in isolation)	"annual_income": "100K_250K"
direct	legal_name, address, email, phone, joint owners, beneficiaries	Role token only	"legal_name": "OWNER_1"
sensitive	tax_id, date_of_birth, delivering institution account number	Exists/empty flag only. Never a value, never a token.	"tax_id": {"state": "vaulted", "filled": true}

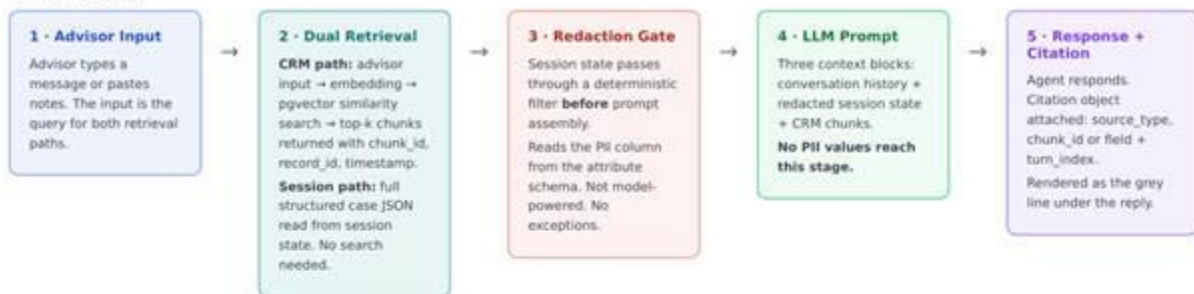
The redaction gate flow:

```
Session State (full values, stored in DB)
|
Redaction Gate (deterministic function)
- none, indirect --> pass through
- direct          --> replace with role token
- sensitive       --> replace with filled/empty flag
|
Prompt Context (safe, no PII reaches the LLM)
```

Citation & Redaction Design Flow

How data reaches the LLM, how citations are generated, and where PII is stripped.

RETRIEVAL PIPELINE



PII REDACTION TIERS

none	Full value passed → account.type, risk_tolerance, funding.method
indirect	Full value passed → income.band, net_worth.band, employment.status
direct	Role token only → legal_name → OWNER_1, email → EMAIL_1
sensitive	Filled/empty flag only → tax_id → ("state":"vaulted","filled":true)

```
// What the LLM actually sees for one attribute
{"field": "applicant.legal_name",
 "redacted_value": "OWNER_1",
 "pii_class": "direct",
 "state": "extracted_confirmed",
 "turn_index": 1}
```

```
// What the LLM sees for a vaulted field
{"field": "applicant.tax_id",
 "state": "vaulted",
 "pii_class": "sensitive",
 "filled": true}
```

CITATION PERSISTENCE

Citation and redaction design flow — from advisor input through dual retrieval, PII redaction, prompt assembly, to cited response. No PII value crosses the redaction gate.

Prompt composition per turn

On every turn, the agent prompt receives three context blocks:

```
[SYSTEM CONTEXT]
|-- conversation_history : last N turns of dialogue
|-- session_state       : redacted case JSON (32 attributes + states)
+-- crm_context         : top-k RAG chunks relevant to this turn
```

The agent reasons across all three but cites differently depending on which source it used.

Citation format

One citation per response, rendered as the grey line below the agent's reply. Two shapes, keyed on source_type.

CRM retrieval citation

```
{
  "citation": {
    "source_type": "crm_retrieval",
    "chunk_id": "crm-note-0417-c3",
    "record_type": "meeting_note",
    "record_id": "NOTE-2026-0417",
    "record_timestamp": "2026-04-17T14:30:00Z",
    "excerpt": "Client mentioned retirement in 3-5 years"
  }
}
```

↳ CRM note from 17 Apr 2026 · “Client mentioned retirement in 3-5 years”

Example: CRM retrieval in action

The advisor asks about existing client information. The agent retrieves a meeting note from the CRM, proposes a value, and cites the exact source.

ADVISOR

What do we know about this client from the CRM?

CONCIERGE

From the April meeting note: they mentioned retirement in 3 to 5 years and wanted something conservative. That lines up with a 5-to-10-year horizon. I have proposed it below; confirm or edit before it is saved.

↳ ↳ CRM note from 17 Apr 2026 · chunk crm-note-0417-c3 · “Client mentioned retirement in 3-5 years, conservative”

AGENT PROPOSES • NOTHING SAVED UNTIL THE ADVISOR CONFIRMS

profile.time_horizon **5_10Y**

[Confirm all] [Edit] [Reject]

New Account

Concierge

Case CASE-0001

ACCOUNT SETUP

Account type

Not set yet

Account purpose

RETIREMENT

confirmed

INVESTMENT PROFILE

Time horizon

5_10Y

proposed

Source: CRM meeting note, 17 Apr 2026

Value proposed from a CRM note. The citation line under the agent reply identifies the exact source. The advisor must confirm before it is saved.

ADVISOR

What do we know about this client from the CRM?

CONCIERGE

From the April meeting note: they mentioned retirement in 3 to 5 years and wanted something conservative. That lines up with a 5-to-10-year horizon. I have proposed it below; confirm or edit before it is saved.

↳ CRM note from 17 Apr 2026 - Chunk crm-note-0417-c3 - "Client mentioned retirement in 3-5 years, conservative"

PROPOSED - CONFIRM BEFORE SAVING

profile.time horizon

5_10Y

Confirm all

Edit

Reject

Type a reply...

Mockup 5.1 — CRM retrieval citation. The agent proposes a time horizon from a meeting note, citing chunk ID and timestamp. The advisor must confirm before it is saved.

Session recall citation

```
{
  "citation": {
    "source_type": "session_recall",
    "field": "financial.annual_income",
    "state": "extracted_confirmed",
    "turn_index": 7,
    "confirmed_at": "2026-09-21T10:04:12Z",
    "value_disclosed": true
  }
}
```

↳ from your answer at step 7 · confirmed at 10:04

Example: session recall in action

The advisor asks about a cross-check. The agent references values confirmed earlier in the same session, citing the exact turn indices.

ADVISOR

The funding amount seems high relative to their profile. Is that an issue?

CONCIERGE

You confirmed \$450,000 at step 11, but the net worth band you gave at step 9 was 250k–1M. The funding amount is within that band, so there is no mismatch. If the client's situation has changed, update the net worth band and I will re-run the cross-check.

↳ ↳ from your answers at steps 9 and 11 · both confirmed · cross-check passed

New Account

Concierge

Case CASE-0001

EMPLOYMENT AND FINANCES

Annual income

100K_250K

Confirmed by advisor at step 7

confirmed

Liquid net worth

250K_1M

Confirmed by advisor at step 9

confirmed

FUNDING

Initial amount

\$450,000

confirmed

The agent recalls what was already confirmed earlier in the session, not from the CRM. The citation references the turn index and confirmation time.

ADVISOR

The funding amount seems high relative to their profile. Is that an issue?

CONCIERGE

You confirmed \$450,000 at step 11, but the net worth band you gave at step 9 was 250k-1M. The funding amount is within that band, so there is no mismatch. If the client's situation has changed, update the net worth band and I will re-run the cross-check.

From your answers at steps 9 and 11 - both confirmed - cross-check passed

Type a reply...

Submit blocked - 4 required fields empty

Mockup 5.2 — session recall citation. The agent references previously confirmed values by turn index and confirmation time, without re-retrieving from the CRM.

PII-safe citation

When the agent cites a direct or sensitive field from session recall, the citation carries the field name and state but never echoes the value:

```
{
  "citation": {
    "source_type": "session_recall",
    "field": "applicant.tax_id",
    "state": "vaulted",
    "turn_index": 4,
    "value_disclosed": false
  }
}
```

↳ tax ID is on file (captured at step 4)

Example: PII-safe citation in action

The advisor asks about a sensitive field. The agent confirms it is on file but never reveals the value, because it was never sent to the LLM.

ADVISOR

Is the tax ID already on file?

CONCIERGE

Yes, the tax ID was captured at step 4 and is vaulted. I cannot see or repeat it. If you need to verify it, check the secure field on the form directly.

↳ ↳ tax ID is on file (captured at step 4) • value not disclosed to agent

ADVISOR

And the date of birth?

CONCIERGE

Also on file, captured at step 5. Same rule: I know it is filled but I have no access to the value itself.

↳ ↳ date of birth on file (captured at step 5) • value not disclosed to agent

New Account • Concierge Case CASE-0001

APPLICANT IDENTITY

Legal name
OWNER_1 from CRM

Tax ID

Agent cannot see this value vaulted

Date of birth
//****
Agent cannot see this value vaulted

Sensitive fields show masked placeholders. The agent knows they are filled but never receives the actual values. The citation confirms presence without disclosure.

ADVISOR

Is the tax ID already on file?

CONCIERGE

Yes, the tax ID was captured at step 4 and is vaulted. I cannot see or repeat it. If you need to verify it, check the secure field on the form directly.

↳ tax ID is on file (captured at step 4) - value not disclosed to agent

ADVISOR

And the date of birth?

CONCIERGE

Also on file, captured at step 5. Same rule: I know it is filled but I have no access to the value itself.

↳ date of birth on file (captured at step 5) - value not disclosed to agent

Type a reply...

Submit blocked - 2 required fields empty

Mockup 5.3 — PII-safe citation. The agent confirms presence of vaulted fields without disclosing values. The citation carries value_disclosed: false.

Persistence on the case record

Citation metadata travels with the field value on the case record. When a case is exported, every value carries its source. This adds roughly 80 bytes per field but means compliance can trace any value back to its origin without reconstructing the chat transcript.

```
// field value with citation on the case record
{
  "field": "financial.annual_income",
  "value": "100K_250K",
  "state": "extracted_confirmed",
  "citation": {
    "source_type": "session_recall",
    "turn_index": 7,
    "confirmed_at": "2026-09-21T10:04:12Z"
  }
}
```

Freshness, not versioning

CRM citations carry the source record's timestamp so the advisor can gauge how fresh the data is. There is no formal version registry for the prototype. If the firm's suitability policy or enum sets change, that is a system-level config change stamped on the case header when the case is opened, not tracked per citation.

KNOWN DESIGN BOUNDARY

The agent cannot compare PII values against each other because it never receives them. When a match check is needed, for example whether the name on the transfer statement matches the account name, the agent flags it for the advisor to verify visually. It can tell the advisor the check is needed. It cannot perform the check itself.

Build notes

Complexity	Low to medium. The redaction gate is a deterministic map over the PII column. The citation format is fixed JSON.
Build order	Build alongside the session state object in week 1. The redaction gate is a prerequisite for any LLM call that receives session context.
Minimum build	Redaction gate covering all four PII tiers, CRM chunk retrieval with citation, session recall citation, citation line rendered in the chat UI, citation persisted on field values.
Before you start	Finalize the PII column in the attribute catalog. Any misclassification passes the wrong data tier to the LLM.

6. Data structure

Thirty-two attributes, twelve groups, five registrations. Sized for a prototype that still covers every shape a real account form contains.

Material and non-material attributes

Every attribute in the catalogue below is marked material or non-material. The split is not about how important a field feels — it is about what an error in it costs to fix after the packet has been submitted.

Class	Definition	Cost of an error after submission
Material	An error changes a regulatory outcome, the title or ownership of the account, or where assets are directed.	Re-papering, re-approval, or a corrected filing. The packet is returned.
Non-material	Supports the workflow but does not itself feed a	Corrected in place. Nothing is re-signed and

	regulatory determination, the account title, or the direction of assets.	nothing is re-filed.
--	--	----------------------

Why these five registrations

Registration	What it exercises
Individual	The baseline path.
Joint with right of survivorship	More than one owner — a repeating block, and more than one signature.
Traditional IRA	Beneficiaries, and percentages that must total exactly 100.
Revocable Trust	An entity rather than a person, with roles and a required document. The most common rejection here is one trustee's signature being missed.
UTMA	An adult controlling a minor's account — and the naming clash below.

The state every value carries

State	Meaning	Agent Scope?
captured_direct	Typed by the advisor; Captured by Agent	Yes
Extracted_confirmed	Agent extracted from Agent request, proposed it, advisor confirmed	Yes
extracted_corrected	Agent proposed it, advisor edited it, advisor confirmed	Yes
prefilled_accepted	Extracted or derived from CRM [Leads Table, Notes], non-material	Yes
prefilled_confirmed	Extracted or derived from CRM [Leads Table, Notes], material, advisor confirmed	Yes
extracted_unconfirmed	Agent proposed it, nobody has confirmed	No — blocks submission
vaulted	Typed into a secure field; the record holds a reference only	Yes

The full attribute list

Account setup

Attribute	Type	Required	Materiality	PII	Note
account.type	enum	always	Material	none	Five registrations. Drives everything else.
account.advisory_program	enum	always	Material	none	Fee schedule and which rules apply.
account.purpose	enum	always	Non-material	none	Free speech is mapped to one option.

Applicant identity

Attribute	Type	Required	Materiality	PII	Note
<code>applicant.legal_name</code>	composite	always	Material	direct	Must match the ID exactly.
<code>applicant.date_of_birth</code>	date	always	Material	direct	Secure field. The agent never sees it.
<code>applicant.tax_id</code>	masked	always	Material	sensitive	Secure field. The agent never sees it.
<code>applicant.citizenship_status</code>	enum	always	Material	indirect	Decides which tax form is generated.
<code>applicant.legal_address</code>	address	always	Material	direct	No PO box. State affects spousal rules.
<code>applicant.email</code>	email	always	Non-material	direct	Where signing requests are sent.
<code>applicant.mobile_phone</code>	phone	always	Non-material	direct	Used for the security code at signing.

Employment and finances

Attribute	Type	Required	Materiality	PII	Note
<code>employment.status</code>	enum	always	Material	indirect	Opens the employer question.
<code>employment.employer_name</code>	text	conditional	Material	indirect	Only if employed or self-employed.
<code>financial.annual_income</code>	enum	always	Material	indirect	Banded, never an exact figure.
<code>financial.liquid_net_worth</code>	enum	always	Material	indirect	Checked against the funding amount.

Household

Attribute	Type	Required	Materiality	PII	Note
<code>household.dependants</code>	integer	optional	Non-material	indirect	Context for planning. Not on the custodian form.
<code>household.minor_dependants</code>	integer	optional	Non-material	indirect	Drives the education-funding prompt and UTMA suggestion.

Investment profile

Attribute	Type	Required	Materiality	PII	Note
<code>profile.investment_objective</code>	enum	always	Material	none	Limits which portfolios are allowed.
<code>profile.risk_tolerance</code>	enum	always	Material	none	The most important cross-check in the set.
<code>profile.time_horizon</code>	enum	always	Material	none	Short horizon plus aggressive objective is flagged.

Regulatory questions

Attribute	Type	Required	Materiality	PII	Note
<code>reg.finra_affiliation</code>	boolean	always	Material	indirect	Yes means an approval letter is required first.
<code>reg.control_person</code>	boolean	always	Material	indirect	Yes means the company ticker is captured.

Joint owners

Attribute	Type	Required	Materiality	PII	Note
<code>joint_owners[]</code>	repeating 1–3	conditional	Material	direct	Each owner answers the identity questions and signs.

Trust and UTMA

Attribute	Type	Required	Materiality	PII	Note
<code>entity.legal_title</code>	text	conditional	Material	indirect	Must match the certification document word for word.
<code>entity.responsible_parties[]</code>	repeating 1–4	conditional	Material	direct	Trustees, or exactly one UTMA custodian (a person).
<code>entity.certification_doc</code>	file	conditional	Material	direct	Blocks submission until uploaded.

Beneficiaries

Attribute	Type	Required	Materiality	PII	Note
<code>beneficiaries[]</code>	repeating 1–6	conditional	Material	direct	Primary shares must total exactly 100.

Trusted contact

Attribute	Type	Required	Materiality	PII	Note
<code>trusted_contact</code>	composite	always	Non-material	direct	Can be declined, but the decline is recorded.

Funding

Attribute	Type	Required	Materiality	PII	Note
<code>funding.method</code>	enum	always	Material	none	Opens the delivering-institution section.
<code>funding.initial_amount</code>	currency	always	Material	indirect	Checked against declared net worth.
<code>funding.delivering_institution</code>	composite	conditional	Material	sensitive	Registration must match or the transfer bounces.

Disclosures

Attribute	Type	Required	Materiality	PII	Note
<code>disclosure.form_crs_ack</code>	attestation	always	Material	none	Exact approved wording only.
<code>disclosure.advisory_agreement_ack</code>	attestation	always	Material	none	Exact approved wording only.

Enum values

Closed sets. The agent maps free speech onto one of these; it never invents a value outside the set, and the compiler rejects anything that is not listed here.

<code>account.type</code>	INDIVIDUAL JOINT_WROS TRADITIONAL_IRA REVOCABLE_TRUST UTMA
<code>account.advisory_program</code>	ADVISOR_DRIVEN UMA THIRD_PARTY_MF DIGITAL_ADVICE
<code>account.purpose</code>	RETIREMENT EDUCATION WEALTH_ACCUMULATION INCOME ESTATE_PLANNING
<code>citizenship_status</code>	US_CITIZEN US_RESIDENT_ALIEN NON_RESIDENT_ALIEN
<code>employment.status</code>	EMPLOYED SELF_EMPLOYED RETIRED STUDENT NOT_EMPLOYED
<code>annual_income</code>	UNDER_50K 50K_100K 100K_250K 250K_500K OVER_500K
<code>liquid_net_worth</code>	UNDER_50K 50K_250K 250K_1M 1M_5M OVER_5M
<code>investment_objective</code>	CAPITAL_PRESERVATION INCOME GROWTH_AND_INCOME GROWTH SPECULATION
<code>risk_tolerance</code>	CONSERVATIVE MOD_CONSERVATIVE MODERATE MOD_AGGRESSIVE AGGRESSIVE
<code>time_horizon</code>	UNDER_3Y 3_5Y 5_10Y OVER_10Y
<code>funding.method</code>	ACAT_TRANSFER ACH CHECK WIRE IRA_ROLLOVER
<code>responsible_party.role</code>	TRUSTEE SUCCESSOR_TRUSTEE UTMA_CUSTODIAN MINOR
<code>beneficiary.class</code>	PRIMARY CONTINGENT

6 · How data moves through the system

WIP – 09/18

7 · Sample case record

The complete stored shape for one in-progress case. Every value carries its state, its source, and where relevant the words it came from. Secure fields hold a reference, never the value. No client-identifying data appears — role tokens are used throughout.

```
{
  "case_id":          "CASE-0001",
  "created_at":       "2026-09-12T09:14:00Z",
  "updated_at":       "2026-09-17T10:48:00Z",
  "catalog_version":  "nao-cat-2026.09.1",
  "registration":     "JOINT_WROS",
  "stage":            "intake",
  "assigned_to":      "ADV_1042",

  "values": {
    "account.type": {
      "value": "JOINT_WROS",
      "state": "extracted_confirmed",
      "heard": "open it jointly with my wife",
      "confidence": 0.94,
      "confirmed_by": "ADV_1042",
      "confirmed_at": "2026-09-17T10:31:00Z"
    },
    "account.advisory_program": {
      "value": "ADVISOR_DRIVEN",
      "state": "captured_direct"
    },
    "applicant.legal_name": {
      "value": { "first": "OWNER_1_FIRST", "middle": null,
                  "last": "OWNER_1_LAST", "suffix": null },
      "state": "prefilled_confirmed",
      "source": "crm.contact",
      "source_age_days": 3
    },
    "applicant.tax_id": {
      "value": "VAULT_REF_02",
      "state": "vaulted",
      "format_valid": true
    },
    "applicant.date_of_birth": {
      "value": "VAULT_REF_01",
      "state": "vaulted"
    },
    "applicant.legal_address": {
      "value": { "line1": "...", "line2": null, "city": "...",
                  "state": "TX", "postal_code": "...", "country": "US" },
      "state": "prefilled_confirmed",
      "source": "crm.contact",
      "source_age_days": 412,
      "flags": ["SOURCE_STALE"]
    },
    "profile.risk_tolerance": {
      "value": "MODERATE",
      "state": "extracted_unconfirmed",
      "heard": "risk is moderate",
      "confidence": 0.95,

```

```

    "blocks_submission": true
  },
  "funding.method": {
    "value": "ACAT_TRANSFER",
    "state": "extracted_confirmed",
    "heard": "moving it over from somewhere else",
    "confidence": 0.79
  },
  "funding.initial_amount": {
    "value": 450000,
    "state": "extracted_corrected",
    "heard": "about 400k",
    "original_value": 400000,
    "corrected_by": "ADV_1042",
    "note": "client added to it after we spoke",
    "flags": ["EXCEEDS_DECLARED_NET_WORTH_BAND"]
  },
  "funding.delivering_institution": {
    "value": {
      "institution_name": "THIRD_PARTY_BROKER",
      "account_number": "VAULT_REF_03",
      "registration_type": "JOINT_WROS",
      "registration_id": null,
      "transfer_scope": "FULL"
    },
    "state": "extracted_unconfirmed",
    "source": "upload.transfer-statement.pdf#page=1",
    "checks": [ { "rule": "acat_registration_match", "status": "pass" } ]
  }
},

"repeating": {
  "joint_owners": [
    { "index": 0, "same_as": "applicant" },
    { "index": 1,
      "legal_name": { "first": "OWNER_2_FIRST", "last": "OWNER_2_LAST" },
      "tax_id": "VAULT_REF_04",
      "state": "captured_direct" }
  ],
  "beneficiaries": [],
  "entity.responsible_parties": []
},

"documents": [
  { "type": "GOV_ID_FRONT", "status": "accepted",
    "checks": ["readable", "not_expired"] },
  { "type": "GOV_ID_BACK", "status": "outstanding",
    "requested_at": "2026-09-16T14:02:00Z" },
  { "type": "ADDRESS_PROOF", "status": "rejected",
    "reason": "older than 90 days", "document_age_days": 243 },
  { "type": "TRUSTEE_CERT", "status": "not_applicable" }
],

"sections": {
  "open": ["joint_owners", "funding.delivering_institution"],
  "closed": ["employment", "entity", "beneficiaries"]
},

"submission": {
  "can_submit": false,
  "unconfirmed_count": 2,
  "missing_required": ["trusted_contact",
    "disclosure.form_crs_ack",
    "disclosure.advisory_agreement_ack"],
  "blocking_documents": ["GOV_ID_BACK"],
  "open_flags": ["EXCEEDS_DECLARED_NET_WORTH_BAND"]
}

```

```
},  
  
"chase": {  
  "status": "active",  
  "days_open": 5,  
  "blocked_on": "CLIENT",  
  "last_contact_at": "2026-09-16T14:02:00Z",  
  "next_action_at": "2026-09-18T09:00:00Z",  
  "next_action": "ESCALATE_TO_ADVISOR",  
  "preferred_channel": "SMS",  
  "messages_sent": 2  
},  
  
"audit": {  
  "turns": 7,  
  "escalations": [  
    { "turn": 6, "kind": "ADVICE_QUESTION", "at": "2026-09-17T10:42:00Z" }  
  ],  
  "model_version": "<pinned at build time>",  
  "prompt_version": "intake-v3"  
}  
}
```